

Name of Student: Mritunjay Pandey

Scholar No.: 23V02078

# INDIAN INSTITUTE OF INFORMATION TECHNOLOGY BHOPAL

Department of Computer Science & Engineering

Mid-Term Examination March – 2025

B.Tech, CSE, 4<sup>th</sup> Semester, Section – II

Subject Name/ Code: CSE -224

Time Allotted: 90 min

MM:20

- NOTE: 1. All questions are compulsory.  
2. Mark answer number clearly.

Q.1 (A) Answer the following question in brief.

2

- List core responsibilities of a database-management system with the problems that would arise if the responsibility were not discharged.
- Given a relation schema  $R(A_1, A_2, A_3, A_4)$  having data of 50 students, calculate its degree and cardinality.

Q.1 (B) Consider the two tables T1 and T2 shown in below figure. Show the results of the following operations: 3

a.  $T1 \bowtie_{T1.P = T2.A} T2$

b.  $T1 \bowtie_{T1.Q = T2.B} T2$

c.  $T1 \bowtie_{T1.P = T2.A} T2$

d.  $T1 \bowtie_{T1.Q = T2.B} T2$

e.  $T1 \cup T2$

f.  $T1 \bowtie_{(T1.P = T2.A \text{ AND } T1.R = T2.C)} T2$

TABLE T1

| P  | Q | R |
|----|---|---|
| 10 | a | 5 |
| 15 | b | 8 |
| 25 | a | 6 |

TABLE T2

| A  | B | C |
|----|---|---|
| 10 | b | 6 |
| 25 | c | 3 |
| 10 | b | 5 |

Q.2 (A) Consider a MOVIE database to design an Entity-Relationship diagram in which data is recorded about the movie industry. The data requirements are summarized as follows: 3

- Each movie is identified by title and year of release. Each movie has a length in minutes. Each has a production company, and each is classified under one or more genres (such as horror, action, drama, and so forth). Each movie has one or more directors and one or more actors appear in it. Each movie also has a plot outline. Finally, each movie has zero or more quotable quotes, each of which is spoken by a particular actor appearing in the movie.
- Actors are identified by name and date of birth and appear in one or more movies. Each actor has a role in the movie.
- Directors are also identified by name and date of birth and direct one or more movies. It is possible for a director to act in a movie (including one that he or she may also direct).
- Production companies are identified by name and each has an address. A production company produces one or more movies.

Q.2 (B) Describe three tables that could be used to store data in a social networking system like Facebook. Consider tables for user profiles, friend relationships, and posts. Explain the attributes each table might include and how they interact to manage user information, connections, and content on the platform.

Q.3 (A) How does relational algebra ensure completeness in expressing queries on relational databases? Explain relational algebra in brief with five essential operations. Consider the following relational schemas for a library:

member(memb\_no, name, dob)

books(isbn, title, authors, publisher)

borrowed(memb\_no, isbn, date)

Also write the relational algebra query for the following question: -

- Find the names of members who have borrowed any book published by "McGraw-Hill".
- Find the name of members who have borrowed all books published by "McGraw-Hill".
- Find the name and membership number of members who have borrowed more than five different books published by "McGraw-Hill".
- For each publisher, find the name and membership number of members who have borrowed more than five books of that publisher.
- Find the average number of books borrowed per member. Take into account that if a member does not borrow any books, then that member does not appear in the borrowed relation at all.

Q.4 (A) Explain the importance of constraints in a Relational Database Management System (RDBMS). Discuss how constraints ensure data integrity, maintain consistency, enforce business rules, and prevent invalid or incorrect data from being entered into the database, thereby improving the reliability and accuracy of the stored information.

Q.4 (B) Given the following table with six tuples and five attributes, generate the all possible valid and completely non trivial functional dependencies.

| A  | B   | C  | D  | E  |
|----|-----|----|----|----|
| 42 | abc | CO | P4 | 42 |
| 43 | pqr | IT | P3 | 43 |
| 44 | xyz | CO | P4 | 44 |
| 45 | xyz | IT | P3 | 45 |
| 46 | mno | EC | P2 | 46 |
| 47 | jkl | ME | P2 | 47 |